THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of images formed by two plane mirrors placed at 90 degrees to each other is: 1

(a) 4 **(b)** 3 **(c)** 1 **(d)** 2

2. A shaving mirror is a: 1

(a) spherical lens **(b)** plane mirror **(c)** convex mirror **(d)** concave mirror

3. A ray of light falls on a plane mirror along the normal. The angle of reflection is: 1

(a) 0 degrees **(b)** 45 degrees **(c)** 90 degrees **(d)** 180 degrees

4. A kaleidoscope works on the principle of: 1

(a) multiple reflections **(b)** refraction only
(c) dispersion **(d)** diffraction

5. A convex lens is thicker at the: 1

(a) edges **(b)** centre **(c)** both equally **(d)** only at the rim

6. The image formed by a plane mirror is: 1

(a) real and inverted **(b)** virtual and inverted
(c) real and erect **(d)** virtual and erect

7. A periscope uses: 1

(a) two convex lenses only **(b)** two plane mirrors
(c) a prism only **(d)** one concave mirror

8. A convex lens is also called a: 1

(a) diverging lens **(b)** plane lens **(c)** prism **(d)** converging lens

9. An image that cannot be obtained on a screen is called a: 1

(a) magnified image only **(b)** diminished image only
(c) virtual image **(d)** real image

10. A mirror that curves inwards is called: 1

(a) flat mirror **(b)** concave mirror **(c)** plane mirror **(d)** convex mirror

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define the focal length and radius of curvature of a mirror. 2

12. What is a magnifying glass? Which lens is used? 2

13. List two properties of the image formed by a plane mirror. 2

14. State two uses of concave mirrors. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why we cannot see our image in a wall though it reflects light. 3

16. A concave mirror has a radius of curvature 30 cm. Find its focal length. 3

17. Explain why a convex lens can burn paper. 3

OR

State three characteristics of the image formed by a plane mirror. 3

18. Why are convex mirrors preferred as rear-view mirrors? 3

OR

Why does a concave lens always give an erect and smaller image? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe how a kaleidoscope is made and how it works. 5

OR

Explain, with reasons, how the human eye uses a convex lens. Why do we use concave lenses for myopia and convex lenses for hypermetropia? 5

20. Describe spherical mirrors, their types, the terms associated and their relation f = R/2. 5

OR

Explain why the image formed by a plane mirror appears behind the mirror. Describe the construction of the image. 5

END OF QUESTION PAPER — Total: 40 Marks

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